PCC-CS393 | ACADEMIC YEAR 2026-2027

Budge Budge Institute of Technology

Travel Booking Data Organizer

Representing Fixed Booking Details via Tuples, Immutability Operations & Aggregated Summaries in Python

PROJECT GROUP MEMBERS

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Course Outcomes: CO4 & CO5

Problem Statement & Objectives

⚠ The Problem

Standard list-based storage leaves confirmed ticket data vulnerable to accidental runtime overrides, data corruption, and unsafe modification during multi-step logic.

◎ Core Solution

Write-Once Integrity: Store tickets as 6-element immutable tuples.

Nested Aggregation: Package atomic tuples in single container collections.

Fast Terminal Querying: Instant ID lookups and revenue summaries.

Why Tuples? (Existing vs. Proposed)

Existing (List-Based)

Mutable Vulnerability: Fares & details can be accidentally modified.

High Overhead: Dynamic resizing buffers consume more RAM.

Manual Calculation: Prone to human errors during iteration.

Proposed (Tuple-Based)

Strict Immutability: Confirmed bookings remain write-protected.

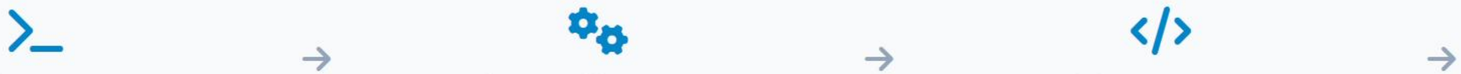
Memory Efficient: Fixed allocation allows faster sequence access.

Automated Routines: Instant revenue accumulation & lookup.

6-Element Tuple Schema & Data Flow

Each passenger record is strictly structured as an immutable 6-field sequence.

Index 0	Index 1	Index 2	Index 3	Index 4	Index 5
Booking ID	Passenger Name	Destination	Travel Date	Transport Mode	Ticket Price



Key System Modules

Linear Search

Matches target ID against tuple index 0 to return exact ticket details instantly.

Revenue Aggregator

Traverses composite tuple collection and sums index 5 for total expenditure.

Route Extractor

Uses tuple comprehension to extract route sequence and counts records via len().

Execution & Test Results

AGGREGATED REVENUE

Rs. 6000

Calculated automatically across all booking tuples

Total Test Cases:	**7 Passed (100%)**
ID Search Accuracy:	**Verified (ID 102 & 999)**
Tuple Operations:	**Indexing & Repetition OK**

Project Conclusion

Successfully satisfies Course Outcomes CO4 & CO5 by proving Python tuples offer a lightweight, immutable alternative for transactional reservation integrity.

Future Scope: JSON File Persistence

Future Scope: Tkinter GUI Interface

Thank You! Questions & Discussion